

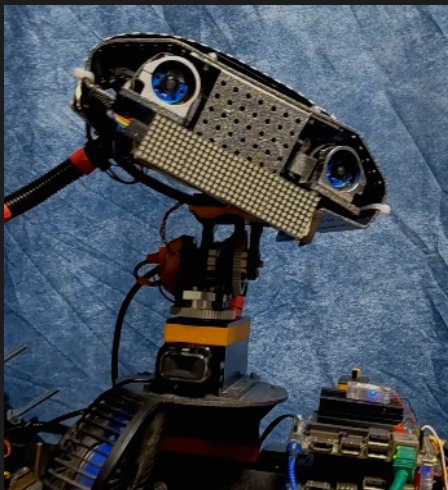
Engineering Roadmap to Phase 1 Alpha Demo

Transitioning from Distributed Emulation to Custom Silicon

[1.0] The Baseline: Phase 0 (Current State)

The current Neurobotics cognitive architecture is successfully running on our proprietary Bubo Emulation Array.

- **Hardware Profile:** The current system relies on a clustered, distributed array of edge-compute nodes to run the complete 7-Tier S-M-R Matrix.
- **Operational Constraints:** While this emulation array successfully proves the software architecture (including Global Workspace Theory and IIT-Phi integration), it requires heavy active cooling, distributed networking, and a massive power supply footprint.
- **The Objective:** To collapse this massive physical footprint into a single, localized \$100mm \times 100mm\$ custom silicon board drawing \$<25W\$ of total system power.



[2.0] Phase 1: Hardware Synthesis & Microchip Integration (Months 1–6)

The first phase of Seed Capital deployment focuses on migrating our validated codebase onto enterprise-grade defense silicon.

- **PolarFire SoC Integration:** We will transition the massively parallelized matrix operations (spatial mapping, vector tracking, and continuous

integrated information metrics) directly into Microchip PolarFire FPGA gate-level logic.

- **PIC64 Execution:** The deterministic, event-driven neural bus and the Aristotelian logic engine will be ported to the Microchip PIC64 architecture.
- **Thermal Profiling:** Initial bench testing to ensure the FPGA fabric can execute complex cognitive loops with zero thermal throttling under heavy load.

[3.0] Phase 2: The SBUD-SB2 Substrate Prototyping (Months 7–12)

With the logic successfully synthesized onto the Microchip development platforms, engineering will shift to custom PCB fabrication.

- **Board Consolidation:** Designing the proprietary SBUD-SB2 (Single Board Brain) substrate to house the entire 7-Tier architecture within the \$100mm \times 100mm\$ form factor.
- **Power Optimization:** Refining the power delivery network to achieve the strict \$<25W\$ operational target, eliminating the need for the heavy fan arrays and ATX power supplies seen in Phase 0.
- **Galvanic Barrier & Security:** Implementing the physical optoisolated barrier and the hardware-enforced Vagus Nerve kill switch directly onto the substrate.

[4.0] Phase 3: The Alpha Demo Deployment (Months 13–18)

The culmination of the Seed round will be the physical demonstration of the SBUD-SB2 board operating a biomechanical chassis.

- **Chassis Integration:** The SBUD-SB2 will be removed from the test bench and mounted into an untethered, mobile biomechanical test platform.
- **Untethered Autonomy:** The Alpha Demo will prove the system's ability to navigate, parse complex environments, and execute deterministic motor actuation with zero reliance on cloud computing, stochastic LLMs, or external server farms.
- **The Deliverable:** A fully autonomous, physical demonstration proving the commercial readiness of the SBUD edge-native architecture, unlocking Series A valuation metrics.

[5.0] Addendum: Phase 3 SOCOM Use-Case (The Tactical SBUD Ecosystem)

The untethered Alpha Demo will concurrently validate the SBUD-SB2 architecture's integration into an autonomous, closed-loop tactical ecosystem designed specifically for DoD/SOCOM medevac and localized cyber intelligence operations.

- **The K-9 Edge Node (Primary Chassis):** Utilizing a heavy-capacity quadruped frame, this unit serves as a mobile edge data center and tactical gear carrier. Powered entirely by the on-board <25W deterministic matrix, it functions as an autonomous, single-operator emergency medevac platform. The K-9 is capable of real-time route calculation and high-speed casualty extraction in completely GPS-denied environments.
- **The 'Arky' (Archaeopteryx) Aerial Scout:** An embedded, chassis-docked secondary aerial drone equipped with radiation-hardened cyber intelligence capabilities. 'Arky' launches autonomously from the K-9 to provide localized aerial overwatch and maintains an encrypted Software-Defined Radio (SDR) link. During an active medevac extraction, 'Arky' remains deployed with the forward SOCOM squad to maintain a continuous cyber intelligence tether while the K-9 physically exfiltrates the casualty.
- **The Smart Litter (Dual-Capacity Towing):** To expand extraction capabilities without acoustic compromise or excessive power drain, the system incorporates an active-assist, gait-matched tethered litter framework. This allows the K-9 platform to simultaneously transport a primary casualty on-board while efficiently towing a secondary casualty or high-value payload over rugged terrain.
- **Operational Mandate:** "No one left behind, and we mean it." Complete tactical resilience under heavy Electronic Warfare (EW) jamming, relying exclusively on bare-metal edge autonomy—zero cloud pings, zero stochastic latency, and instantaneous battlefield execution.